Software Devlopment Life Cycle (SDLC)

1. Requirements

Purpose of the 'requirements' is to Collect and evaluate business needs.

Some of the importance it has is it serves as the project's foundation and ensures stakeholder expectations are met.

Some of the key Activities are to conduct stakeholder interviews, Document requirements and Analyze feasibility.

This phase feeds into Design by providing clear objectives.

2. Design

Purpose of design phase is to devlop system architecture and technical specifications.

The Importance it brings is outlining how the system will function and identifying potential technical solutions, and risks in advance.

It's key Activities that it contributes to SDLC process & Designing UI/UX mockups, creating database schemas, and outling system and softward architecture

3. Implementation

Purpose of Implementation phase is to Engage in coding and devlop the system.

It's significance is that it converts designs into a functioning product and Emphasizes quality coding practises.

It's key activities are ~~promoting~~ performing programming, conducting code reviews and integrating components.

This phrase transitions into Testing to ensure functionality.

4. Testing

It's purpose is to Assess the software's quality and accuracy.

It's significance is to detecting and fixing bugs, and conforming the system meets established requirements.

It's key Activities are Execute unit testing, Conduct integration testing and perform user acceptance testing

5. Deployment

Purpose is to lauch the final product for users

It's importance includes to make the system accessible to end users and involves careful rollout and ongoing monitoring

It's key Activities are to release to production, provide user training and monitor performance

May revert to requirements for enhancements, initiating a new cycle.

The SDLC is an iterative process, feedback at any stage can loop back to refine previous phases.